

What is CSS Multi-column Layout?

CSS Multi-column Layout is a module that allows you to flow content across multiple columns in a single container, similar to newspaper or magazine layouts. Content automatically distributes across columns, and columns break efficiently at logical points. This is perfect for creating responsive text layouts, content galleries, and magazine-style presentations without using CSS Grid or Flexbox.

Browser Support: All modern browsers support CSS Multi-column Layout. Chrome, Firefox, Safari, and Edge all support it. Use `-webkit-` and `-moz-` prefixes for older versions and complete compatibility.

Core Concepts

When to Use Multi-column Layout

- Text-heavy content:** Articles, blog posts, documentation
- Magazine layouts:** Newspaper-style formatting
- Product listings:** Simple grid of items flowing across columns
- Responsive design:** Content adapts to available width
- Image galleries:** Simple flowing layouts
- Comment threads:** Chat-like layouts

Key Differences from Grid/Flexbox

Property Grid Flexbox Multi-column **Direction** 2D (rows & columns) 1D (row or column) Column-based flow **Content Flow** Fixed placement Fixed direction Auto-distributing **Best For** Page layouts Component layouts Text/flowing content **Item Control** Fine-grained Flexion Limited

Multi-column Properties

1. column-count

Specifies the number of columns. Content is divided equally among these columns.

Syntax: column-count: <integer> | auto;

Values:

- auto (default) - Let browser determine based on other properties
- 1 to n - Fixed number of columns

Examples:

```
column-count: 2; /* Two columns */
column-count: 3; /* Three columns */
column-count: auto; /* Dynamic based on column-width */
```

2. column-width

Suggests the optimal column width. Browser calculates how many columns fit.

Syntax: column-width: <length> | auto;

Values:

- auto (default)
- Any length: 200px, 15em, 20ch, etc.

Examples:

```
column-width: 250px; /* Each column ~250px wide */
column-width: 15em; /* Each column ~15em wide */
column-width: 200px; /* Responsive: fits as many as possible */
```

3. columns Shorthand

Combines column-count and column-width in one declaration.

Syntax: columns: [column-count] [column-width];

Examples:

```
columns: 2; /* 2 columns, auto width */
columns: 3 200px; /* 3 columns, 200px each */
columns: auto 200px; /* Auto count, 200px width */
```

4. column-gap

Sets the space between columns.

Syntax: column-gap: <length> | normal;

Default: normal (usually 1em)

Examples:

```
column-gap: 20px; /* 20px space */
column-gap: 2em; /* 2em space */
column-gap: normal; /* Browser default */
```

5. column-rule

Draws a line between columns (like a border between columns).

Syntax: column-rule: [width] [style] [color];

Examples:

```
column-rule: 1px solid #ccc;
column-rule: 2px dashed #667eea;
column-rule: 3px double black;
```

Individual Properties:

```
column-rule-style: solid;
column-rule-width: 2px;
column-rule-color: #667eea;
```

6. column-fill

Controls how content fills columns (height-based or sequential).

Syntax: column-fill: balance | auto;

Values:

- balance (default) - Content distributed to fill all columns equally
- auto - Columns filled sequentially, may leave last column empty

Example:

```
column-fill: auto; /* Fill left-to-right sequentially */
column-fill: balance; /* Distribute evenly */
```

7. column-span

Allows an element to span across multiple columns (or all).

Syntax: column-span: none | all;

Values:

- none (default) - Element stays within one column
- all - Element spans all columns

Example:

```
h2 {
  column-span: all; /* Headings span all columns */
}
p {
  column-span: none; /* Paragraphs in single column */
}
```

8. break-inside

Controls whether elements can break inside columns.

Syntax: break-inside: auto | avoid | avoid-page | avoid-column | avoid-region;

Common Values:

- auto (default) - Element can break
- avoid - Try to keep element within one column
- avoid-column - Never break within column

Example:

```
.card {
  break-inside: avoid; /* Keep cards in single column */
}
```

Interactive Multi-column Demos

2-Column Layout (column-count: 2)

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column-count: 2; column-gap: 20px;

3-Column Layout (column-count: 3)

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column-count: 3; column-gap: 20px;

Auto Columns (columns: auto 200px)

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columns: auto 200px; (Responsive column count)

With Column Rule (column-rule: 2px solid)

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columns: 2; column-rule: 2px solid #667eea;

Cards with break-inside: avoid

Card 1
This card stays in one column and won't break.

Card 2
Each card is kept whole without breaking across columns.

Card 3
Perfect for maintaining card integrity in layouts.

Card 4
Content flows across columns naturally.

Card 5
Cards distribute across available columns.

Card 6
Last card completes the layout.

columns: 3; .card { break-inside: avoid; }

Common Use Cases & Patterns

2-Column Article Layout

```
article {
  column-count: 2;
  column-gap: 30px;
  column-rule: 1px solid #ddd;
}

article h1 {
  column-span: all; /* Title spans both columns */
}
```

Responsive Columns

```
.content {
  columns: auto 250px; /* Auto count, 250px each */
  column-gap: 20px;
}

/* More columns on wider screens, fewer on mobile */
@media (max-width: 768px) {
  .content { columns: auto 350px; }
}
```

Product Grid

```
.product-list {
  columns: auto 200px;
  column-gap: 15px;
}

.product {
  break-inside: avoid; /* Keep products whole */
  margin-bottom: 15px;
}
```

Blog Comment Section

```
.comments {
  columns: auto 200px;
  column-gap: 20px;
}

.comment {
  break-inside: avoid;
  background: #f9f9f9;
  padding: 15px;
  margin-bottom: 15px;
  border-radius: 5px;
}
```

Magazine-Style Layout

```
.magazine {
  columns: 3;
  column-gap: 30px;
  column-rule: 2px solid #667eea;
}

.magazine h1 {
  column-span: all;
  margin-bottom: 20px;
}

.pull-quote {
  column-span: all;
  font-size: 1.5em;
  font-weight: bold;
  margin: 20px 0;
}
```

Gallery with Auto-fit Columns

```
.gallery {
  columns: auto 150px; /* Auto count, ~150px images */
  column-gap: 10px;
}

.gallery img {
  width: 100%;
  height: auto;
  margin-bottom: 10px;
  display: block;
}
```

Advanced Techniques

Spanning with Media Queries

```
/* Desktop: Multi-column */
@media (min-width: 1024px) {
  article {
    columns: 2;
  }
}

/* Tablet: Single column */
@media (max-width: 1023px) {
  article {
    columns: 1;
  }
}
```

Balancing Columns with Height

```
.balanced-columns {
  columns: 2;
  column-fill: balance; /* Default: balance height */
}

.sequential-columns {
  columns: 2;
  column-fill: auto; /* Fill left-to-right */
}

/* With height limit */
.clamped-height {
  columns: 3;
  column-fill: auto;
  height: 500px; /* Limit total height */
  overflow: auto; /* Add scroll if needed */
}
```

Orphans and Widows Control

```
/* Prevent orphaned lines at column bottom */
p {
  orphans: 3; /* At least 3 lines at bottom */
  widows: 3; /* At least 3 lines at top */
}
```

Complex Break Control

```
h2 {
  break-after: avoid; /* No break after heading */
  margin-bottom: 10px;
}

.card {
  break-inside: avoid; /* Keep card whole */
  break-after: auto; /* Allow break after card */
}
```

Browser Compatibility & Prefixes

Chrome/Edge: Full support. Use both `-webkit-` and unprefixed versions.

Firefox: Full support. Use both `-moz-` and unprefixed versions.

Safari: Full support with `-webkit-` prefix. Include both versions.

IE 10/11: Limited support with `-ms-` prefix. Test thoroughly.

Vendor Prefix Strategy

```
.multi-column {
  /* Webkit (Chrome, Safari) */
  -webkit-column-count: 2;
  -webkit-column-gap: 20px;
  -webkit-column-rule: 1px solid #ddd;

  /* Mozilla (Firefox) */
  -moz-column-count: 2;
  -moz-column-gap: 20px;
  -moz-column-rule: 1px solid #ddd;

  /* Standard */
  column-count: 2;
  column-gap: 20px;
  column-rule: 1px solid #ddd;
}
```

Best Practices

- Use column-width for responsive layouts:** `columns: auto 250px` adapts better to screen sizes than fixed column-count.
- Always use vendor prefixes:** Include `-webkit-`, `-moz-`, and standard versions.
- Prevent element breaking:** Use `break-inside: avoid` on cards, images, and important content.
- Test on actual devices:** Multi-column behavior varies slightly between browsers.
- Combine with media queries:** Adjust column count for different screen sizes.
- Set reasonable gaps:** Too small gaps make columns feel cramped; too large wastes space.
- Use column-span sparingly:** Too many spanning elements disrupt column flow.
- Consider readability:** Too many narrow columns reduce readability. 2-3 columns is usually optimal.
- Test with real content:** Empty placeholders don't show layout issues. Use actual text/images.

Common Mistakes to Avoid

- Mixing column-count and column-width:** Use one or the other, not both as separate rules (use `columns` shorthand instead).
- Forgetting break-inside:** Cards and images will break awkwardly without this.
- Not testing with vendor prefixes:** Older browsers may not recognize unprefixed properties.
- Too narrow columns:** Single-word lines are hard to read. Keep minimum column width reasonable (200px+).
- Excessive column-span usage:** Overusing `column-span: all` destroys column layout flow.
- Not considering content hierarchy:** Important content may be buried in later columns.
- Fixed height without overflow handling:** Content overflow with `height: 200px` is invisible.
- Ignoring mobile:** Multi-columns look cramped on small screens. Use media queries to disable.
- Not testing with varied content:** Short text looks fine; long text may break unexpectedly.

Multi-column vs Grid vs Flexbox

Feature	Multi-column	Grid	Flexbox
Use Case	Text/flowing content	Page layout	Component layout
Dimensions	1D (columns)	2D	1D
Content Flow	Auto-distributing	Fixed placement	Fixed direction
Item Placement	Automatic	Explicit	Automatic
Responsive	Built-in (auto columns)	Media queries	Media queries
Browser Support	Excellent (with prefixes)	Excellent	Excellent

Quick Reference Guide

Template: Fixed 2-Column

```
.container {
  -webkit-column-count: 2;
  -moz-column-count: 2;
  column-count: 2;
  -webkit-column-gap: 20px;
  -moz-column-gap: 20px;
  column-gap: 20px;
}
```

Template: Responsive Auto Columns

```
.container {
  -webkit-columns: auto 250px;
  -moz-columns: auto 250px;
  columns: auto 250px;
  -webkit-column-gap: 20px;
  -moz-column-gap: 20px;
  column-gap: 20px;
}
```

Template: Magazine Layout

```
.magazine {
  -webkit-column-count: 3;
  -moz-column-count: 3;
  column-count: 3;
  -webkit-column-gap: 30px;
  -moz-column-gap: 30px;
  column-gap: 30px;
  -webkit-column-rule: 2px solid #ddd;
  -moz-column-rule: 2px solid #ddd;
  column-rule: 2px solid #ddd;
}

.magazine h1 {
  -webkit-column-span: all;
  -moz-column-span: all;
  column-span: all;
}
```

Template: Card Grid

```
.cards {
  -webkit-columns: auto 200px;
  -moz-columns: auto 200px;
  columns: auto 200px;
  -webkit-column-gap: 15px;
  -moz-column-gap: 15px;
  column-gap: 15px;
}

.card {
  -webkit-break-inside: avoid;
  -moz-break-inside: avoid;
  break-inside: avoid;
  margin-bottom: 15px;
}
```